Java Object-Oriented Programming

Comprehensive Assignment Questions & Core OOP Exercises

ENCAPSULATION

1. Write a Java program to create a class called `Smartphone` with private instance variables `brand`, `model`, and `storageCapacity`. Provide public getter and setter methods to access and modify these variables. Add a method called `increaseStorage()` that takes an integer value and increases the `storageCapacity` by that value.

2. Write a Java program to create a class called `ElectronicsProduct` with attributes for product ID, name, and price. Implement methods to apply a discount and calculate the final price. Create a subclass `WashingMachine` that adds a warranty period attribute and a method to extend the warranty.

3. Write a Java program to create a class called `Library` with a collection of books and methods to add and remove books.

INHERITANCE & CLASS HIERARCHY

4. Write a Java program to create a class called `CustomerOrder` with attributes for order ID, customer name, and order date. Create a subclass `OnlineOrder` that adds attributes for delivery address and tracking number. Implement methods to calculate delivery time based on the address and update the tracking status.

5. Write a Java program to create a class called `Reservation` with attributes for reservation ID, customer name, and date. Create subclasses `ResortReservation` and `RailwayReservation` that add specific attributes like room number for resorts and seat number for trains. Implement methods to check reservation status and modify reservation details.

6. Write a Java program that creates a class hierarchy for employees of a company. The base class should be `Employee`, with subclasses `Manager`, `Developer`, and `Programmer`. Each subclass should have properties such as name, address, salary, and job title. Implement methods for calculating bonuses, generating performance reports, and managing projects.

POLYMORPHISM & METHOD OVERRIDING

7. Write a Java program to create a base class `Vehicle` with methods `startEngine()` and `stopEngine()`. Create two subclasses, `Car` and `Motorcycle`. Override the `startEngine()` and `stopEngine()` methods in each subclass to start and stop the engines differently.

8. Write a Java program to create a class called `BankAccount` with methods `deposit()` and `withdraw()`. Create a subclass called `SavingsAccount` that overrides the `withdraw()` method to prevent withdrawals if the account balance falls below one hundred.

9. Write a Java program to create a class called `Vehicle` with a method called `drive()`. Create a subclass called `Car` that overrides the `drive()` method to print "Repairing a car."

ABSTRACTION

10. Write a Java program to create an abstract class `Shape3D` with abstract methods `calculateVolume()` and `calculateSurfaceArea()`. Create subclasses `Sphere` and `Cube` that extend the `Shape3D` class and implement the respective methods to calculate the volume and surface area of each shape.

COMPREHENSIVE OOP PROJECT

11. Design a Java program for an **Employee Management System** that demonstrates the concepts of Encapsulation, Inheritance, and Polymorphism.

- Create a base class `Employee` with private data members `id` and `name`, a constructor, getter methods, an overridable `calculateSalary()` method, and a method to display details.
- Create two derived classes:
  - `FullTimeEmployee`: adds `monthlySalary` and overrides `calculateSalary()`.
  - `PartTimeEmployee`: adds `hoursWorked` & `hourlyRate` and overrides `calculateSalary()`.
- In `main()`: instantiate objects using a parent class reference, display details, and demonstrate runtime polymorphism.